Pavithraa Satchithanantham

psatchit@asu.edu | LinkedIn | Portfolio Website | GitHub

AI Engineer with 2+ years of experience with proficiency in agentic AI, RAG, LLM, computer vision and ML deployments. Architected and built an award-winning agentic RAG platform on AWS (Lambda, Bedrock, OpenSearch) powered by Claude Sonnet 3.7 improving document retrieval efficiency from 23% to 78%, serving 50+ daily engineers. Published two international IEEE research papers.

EDUCATION

Masters of Science, Computer Science **(GPA: 3.93)** — January 2024 – December 2025
Arizona State University, ***New American University Scholarship Awardee ($8000)*** — *Tempe, Arizona*

Bachelors of Engineering, Electronics and Communication Engineering **(GPA: 3.72)** — August 2019 – May 2023
M.S. Ramaiah Institute of Technology — *Bengaluru, India*

SKILLS

Programming Languages: Python, Java, C++, Javascript, SQL, C Programming
Machine Learning/ Data Science: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, Seaborn, Matplotlib
AI Specializations: Agentic AI, Generative AI, LLM, NLP, MLOps, CNN, RNN, LSTM, MLP, Computer Vision, Deep Learning, Regression, Decision Trees, Linear Algebra, Probability, Transformer based Large Language Models, OpenAI, LangChain, vector databases
Platforms & Cloud: AWS (Lambda, S3, EC2, ECR, Bedrock, Knowledge Bases, OpenSearch, SQS, SNS), Docker, Git, MySQL, MongoDB
Frameworks: React.js, Next.js, Node.js, REST API, Unreal Engine, PowerBI

PROFESSIONAL EXPERIENCE

AI Engineering Intern | EchoStar Corporation (Dish Network) | *Denver, Colorado* — June 2025 – August 2025

- Architected and built end-to-end proprietary advanced agentic AI Retrieval augmented Generation (RAG) system that boosts team productivity by roughly 45% and increases document retrieval efficiency from 23% to 78%
- Designed serverless Lambda orchestration architecture powered by Claude Sonnet 3.7 (custom metadata filtering, hybrid BM25 + text-embedding Amazon TTE over OSS vectorDB) serving 50+ daily engineers
- Improved top-5 retrieval MRR 47% from initial 8% and cut p95 latency from 8s to 920ms via re-ranking & hierarchical chunking
- Innovated adaptive, role aware 'Core' (Developer) and 'Nexus' (Executive) personas using system prompting + inference tuning (temperature, top_p, top_k, max_tokens) to tailor grounded responses

Machine Learning Engineer | M.S. Ramaiah Institute of Technology | *Bengaluru, India* — March 2022 – April 2023

- Designed a video based intrusion detection framework, built and benchmarked two novel architectures ConvLSTM2D + MaxPool3D (~50k params) model against LRCN (~500k params) on 20 frame 64x64 clips, with 84.21% and 94.74% accuracy
- Deployed the ConvLSTM detector on an NVIDIA Jetson Nano (4GB, 5W mode) for real time perimeter surveillance by ~7 clips/s throughput at ~150ms latency per 20 frame clip within an 800MB RAM footprint

AI Engineer | Curl | *Bengaluru, India* — October 2021 – February 2022

- Architected an end-to-end autonomous art pipeline converting park camera images into stylized artworks via a StyleGAN2-ADA + AdaIN model, minted them as NFTs and sole on OpenSea
- Fine-tuned StyleGAN2-ADA from an FFHQ pretrained checkpoint on a curated 4200 image nature corpus using a single Colab Pro A100 (40 GB) and reached FID ~22 in ~15 GPU hours

AWARDS & HONORS

- **Winner**, **CPAW Curiosity award** for exceptional curiosity and novelty throughout the internship, Aug 2025
- **Winner**, **(National) Smart India Hackathon 2022**, Ministry of Education, Govt. of India, cash prize of $1200, Aug 2022
- **Runner up, (International) Sovereign Nature Initiative Winter Hackathon Challenge**, cash prize and tokens, Feb 2022

PROJECT EXPERIENCE

Drone Detection and Tracking System | MSRIT in colab with **DRDO |** *Bengaluru, India* — October 2021 – June 2022

- Boosted real time small drone detection on sub-20 px targets by +5 pp mAP@0.5 on a custom 2k image drone vs bird dataset, by adding a P2/stride-4 detection head to YOLOv5l architecture increasing mAP@0.5 from 87.3% to 89.3%
- Recomputing dataset specific anchors via k-means + genetic algorithm refinement increasing mAP@0.5 from 89.3% to 94.2%
- Engineered monocular threat-assessment heuristic module for drones during a 36 hour hackathon by using bounding box as a proxy for range estimation by computing a smoothed dA/dt signal to classify drone radial motion into 3 states (approach, recede, hover/lateral) with 90% classification accuracy on test sequences

PUBLICATIONS & RESEARCH

- GARUDA: Third Eye for Detecting and Tracking Drones | **2023 IEEE ICDDS**
- A Deep Learning Approach to Classify Drones and Birds | **2022 IEEE Mysuru Sub Section International Conference**